

Plant Disease Detection Flask Application Report

Introduction

The Plant Disease Detection Flask Application is an AI-based web application developed to identify diseases in plant leaves through image analysis. Agriculture plays a vital role in food production, and plant diseases can greatly affect crop quality and productivity. Traditional disease identification methods require expert knowledge and consume significant time. This project provides an automated and user-friendly solution that allows users to upload plant leaf images and receive instant disease predictions. The application combines machine learning, computer vision, and web technologies to support smart agriculture practices.

Abstract

The project focuses on developing a deep learning-based system for detecting plant diseases from leaf images. A convolutional neural network model was trained using plant disease datasets containing healthy and infected leaf samples. The trained model was integrated into a Flask web application where users can upload images for prediction. OpenCV and NumPy were used for image preprocessing, while HTML templates were used to create an interactive interface. The system demonstrates how artificial intelligence can improve agricultural monitoring and disease management.

Tools Used

- Python – Used as the main programming language for backend development and machine learning.
- Flask Framework – Used to build the web application and handle routing and predictions.
- TensorFlow/Keras – Used for training and loading the deep learning model.
- OpenCV – Used for image preprocessing such as resizing and normalization.
- NumPy – Used for numerical computations and array handling.
- HTML/CSS – Used for creating the frontend user interface.
- H5 Model File – Used to store the trained plant disease prediction model.

Steps Involved in Building the Project

1. Collected plant leaf image datasets containing healthy and diseased samples.
2. Preprocessed the dataset by resizing images and converting them into numerical arrays.
3. Trained a convolutional neural network model using TensorFlow/Keras.
4. Evaluated model accuracy and saved the trained model in H5 format.
5. Developed a Flask backend to upload images and generate predictions.
6. Used OpenCV to preprocess uploaded images before prediction.
7. Created frontend pages using HTML and CSS for user interaction.
8. Connected the frontend and backend to display disease prediction results.
9. Tested the application with multiple plant leaf images to ensure proper functionality.

Conclusion

The Plant Disease Detection Flask Application successfully demonstrates the integration of artificial intelligence with agriculture. The project provides a fast and efficient method for identifying plant diseases using image classification techniques. By combining deep learning, computer vision, and web development, the application offers an accessible platform for users to detect diseases accurately. The project highlights the practical importance of AI-driven solutions in improving agricultural productivity and reducing crop losses.